

Energy Balance Climate Modelling — A Ten-Lecture Course

Built from: `CLIMATE_MODEL_REPORT.md` (orbital perturbation climate model) and the `orbital_climate/` package.

Course design

Audience. A student with no climate science background. Assumed: first-year calculus (comfortable differentiating, has seen an ODE, may never have solved a PDE), basic Python, no prior orbital mechanics, no numerical methods.

Governing principle. *Never introduce a term before the student has felt the pain it solves.* Every term in the final equation arrives as the answer to a failure the student has personally observed. The final governing equation is not shown in full until Lecture 7.

Deferred-Kepler ordering. Orbital mechanics is a prerequisite for the *scenario*, not for the *climate physics*. Global mean temperature, latitudinal gradients, ice-albedo feedback and the snowball bifurcation are all comprehensible on a circular orbit. Kepler arrives in Lecture 5, once there is a working climate model that needs an orbit to drive it.

Failure-forward. The report's §7 (Corrections and negative results) and §4.2 (the failed heat-capacity blend) are the most pedagogically valuable material in the source document. They are distributed through the course as primary content, not appended as an afterthought.

Format assumption. Ten 90-minute sessions, each roughly 50 min lecture + 40 min lab. Exercises are specified to be completable in the lab slot with a skeleton provided; the "extension" items are homework.

Sequence at a glance

#	Title	Introduces	Report §§
1	Why is Earth 288 K?	Energy balance, albedo, linear OLR	3.1, 3.4
2	Why isn't every latitude the same?	Insolation geometry, obliquity, seasons	2.2
3	Why don't the poles freeze?	Diffusive heat transport	3.1, 3.2
4	Ice, feedback, tipping points	Albedo feedback, bifurcation	3.1, 6.2
5	The orbit isn't a circle	Kepler, eccentricity, Milankovitch	2.1, 2.3
6	Time constants	Heat capacity, two-surface model	4.1–4.6
7	Turning a PDE into a program	Grids, IMEX stepping, conservation	3.2, 3.3
8	How do you know it isn't lying?	Validation hierarchy	2.3, 3.5, 8, Appendix
9	Spurious signals and honest limits	Confounding, regime dependence	6.3, 6.7, 9
10	The full pipeline (capstone)	Frame recovery, sweep results	5, 6

ACT I — What sets a planet's temperature

Lecture 1 — Why is Earth 288 K?

Learning objectives

By the end, the student can:

1. Derive the effective radiating temperature of a planet from first principles and explain where the factor of 4 comes from.
2. State the greenhouse gap (~ 33 K) as an observed discrepancy requiring explanation, not as a mystery.
3. Write down the linear OLR parameterisation $(A + B \cdot T)$ and articulate precisely what kind of approximation it is.
4. Explain why a model can be quantitatively accurate near its calibration point and quantitatively meaningless away from it.

Narrative

Energy in equals energy out; everything else is detail. Absorbed solar $(S_o(1-\alpha)/4)$ balanced against emitted longwave. Sphere-vs-disc geometry gives the 4. Present-day numbers give ~255 K. Observed surface temperature is 288 K. Name the 33 K gap, gesture at the greenhouse mechanism, and then — crucially — decline to model it from physics.

Instead introduce $(OLR = A + B \cdot T)$ as a deliberate empirical cheat: a straight line fitted to satellite observations in the neighbourhood of 288 K, with $(A = 203.3 \text{ W/m}^2)$ and $(B = 2.09 \text{ W/m}^2/\text{°C})$. Say plainly that this is the single largest approximation in the entire model and that Lecture 9 will show it producing confident nonsense.

The student has now built the zero-dimensional version of the whole model. Everything that follows adds structure to this.

Exercise 1 — Zero-dimensional planet

Provide: nothing but a calculator/notebook.

Task:

1. Compute effective temperature for Earth ($S_o = 1361 \text{ W/m}^2$, $\alpha = 0.30$).
2. Repeat for Mars ($S_o = 586$, $\alpha = 0.25$) and Venus ($S_o = 2601$, $\alpha = 0.77$).
3. Compare against observed surface temperatures (Mars ~210 K, Venus ~737 K). Which planet has the largest greenhouse gap, and does the ranking match atmospheric mass?
4. Using $(A + B \cdot T)$ with the given constants, solve for the coalbedo that yields exactly 288.15 K. Compare to the report's calibrated $(a_o = 0.676)$.

Expected result: Step 4 should land within a few thousandths of 0.676; a discrepancy is itself informative (the report's value is calibrated against a *latitudinally resolved* model with a $(a_2 \cdot P_2(x))$ structure, not a global constant).

Discussion prompt: Venus's gap is ~500 K. What does that tell you about extrapolating a linear OLR fit?

Common misconceptions to pre-empt

- "The greenhouse effect traps heat" → it raises the altitude of emission; the planet still radiates to balance.
- The factor of 4 is *not* about day/night.

Lecture 2 — Why isn't every latitude the same?

Learning objectives

1. Derive solar declination from obliquity and solar longitude.
2. Explain the sunset hour angle and why its arccos argument must be clamped.

3. Recognise polar day and polar night as the two saturated branches of a single formula rather than special cases requiring separate code.
4. Predict, before computing, that the summer pole receives more daily-mean insolation than the equator.

Narrative

Pure geometry, no physics. Build up in order:

- solar longitude $(\lambda = v + \lambda_p)$ (with v held at a circular-orbit value for now — the anomaly machinery is deferred)
- declination $(\delta = \arcsin(\sin \varepsilon \cdot \sin \lambda))$
- sunset hour angle $(H_0 = \arccos(-\tan \phi \cdot \tan \delta))$
- daily-mean flux $(Q = (S_0/\pi)/r^2 \cdot (H_0 \sin \phi \sin \delta + \cos \phi \cos \delta \sin H_0))$

Spend real time on the clamp. When $(|-\tan \phi \cdot \tan \delta| > 1)$ the sun either never sets or never rises; clamping to ± 1 yields $(H_0 = \pi)$ and $(H_0 = 0)$ respectively, and the flux formula then gives the right answer on both branches with no branching logic. This is a habit of mind worth naming explicitly: **design the edge case out of existence rather than testing for it.**

Exercise 2 — The insolation surface

Provide: `(orbital_climate/insolation.py)`, circular orbit assumed ($e = 0$).

Task:

1. Produce a filled contour plot of $(Q(\phi, \text{day}))$ over the full year, ϕ from -90° to $+90^\circ$.
2. Mark the latitude and day of maximum daily-mean insolation on the whole plot.
3. Verify the equator-at-equinox value equals (S_0/π) .
4. Set $\varepsilon = 0$ and re-plot. Set $\varepsilon = 90^\circ$ and re-plot.

Expected result: Global maximum sits at the *summer pole*, not the equator — typically a surprise. At $\varepsilon = 0$ the plot is uniform in day. At $\varepsilon = 90^\circ$ the poles become the hottest places on the planet by a wide margin.

Extension: At what obliquity does the annual-mean insolation at the pole first exceed that at the equator? (Answer near 54° ; the report's sweep contains runs at 74° .)

Common misconceptions

- "Seasons are caused by distance from the Sun" — this lecture kills it permanently, since the model here has a circular orbit and still has seasons.
-

Lecture 3 — Why don't the poles freeze and the tropics boil?

Learning objectives

1. Compute local radiative equilibrium temperature at each latitude and identify why it is unphysical.
2. Explain diffusion as a parameterisation of atmospheric and oceanic heat transport, and state honestly what it is *not*.
3. Describe the qualitative effect of varying the diffusion coefficient D across its full range.

Narrative

Start by *running the broken model*. With $D = 0$ and ice suppressed, each latitude independently reaches $T = (\bar{Q} \cdot a_0 - A)/B$. Plot it. The poles are catastrophically cold, the tropics too hot, and the equator-to-pole gradient is roughly double the observed one.

Now introduce the transport term:

$$D \frac{\partial}{\partial x} \left[(1 - x^2) \frac{\partial T}{\partial x} \right] \quad \text{where } x = \sin(\text{latitude})$$

Be explicit that this is a *parameterisation*, not a derivation. Real poleward heat transport is done by baroclinic eddies, the Hadley circulation and ocean gyres; none of that is in here. $D = 0.58 \text{ W/m}^2/\text{°C}$ is a number chosen because it reproduces the observed gradient. This is the second-largest approximation in the model after linear OLR.

Note the geometric factor $(1 - x^2) = \cos^2(\text{latitude})$ vanishing at both poles — this is why the no-flux boundary condition comes for free. Flag it now; Lecture 7 returns to why that matters numerically.

Exercise 3 — The transport dial

Provide: `orbital_climate/ebm.py` with ice-albedo feedback disabled.

Task:

1. Run to equilibrium with $D = 0$. Verify each latitude matches the analytic local-equilibrium formula to within 0.05 °C .
2. Sweep D over $\{0, 0.1, 0.3, 0.58, 1.5, 5.0\}$. Overlay the equilibrium temperature profiles.
3. Plot equator-minus-pole temperature difference against D .
4. For each D , confirm the *global mean* is unchanged.

Expected result: Step 4 is the point of the exercise. Diffusion redistributes energy; it cannot create or destroy it. The global mean is fixed by the radiation balance alone. Students who expected transport to warm the planet have learned something structural.

Tolerance: Global mean should be invariant to $\sim 0.01 \text{ K}$ across all D .

Lecture 4 — Ice, feedback, and tipping points

Learning objectives

1. Explain the ice-albedo feedback loop in words and identify it as positive feedback.
2. Distinguish an *emergent* model behaviour from an *imposed* one.
3. Describe the Budyko–Sellers snowball bifurcation and locate the forbidden temperature gap.
4. Explain why the existence of the bifurcation is robust while its precise location is not.

Narrative

Make albedo temperature-dependent: $a_{\text{ice}} = 0.38$ wherever $T < T_{\text{ice}} = -10\text{ }^{\circ}\text{C}$, otherwise $a_0 + a_2 \cdot P_2(x)$. The model is now nonlinear, and everything interesting follows from that one change.

Walk the feedback loop explicitly: cooling $\rightarrow$ more ice $\rightarrow$ higher albedo $\rightarrow$ less absorbed $\rightarrow$ more cooling. Then run the model down a slow dimming ramp and watch the ice edge march equatorward, slowly at first, then catastrophically.

The central teaching moment: nobody wrote a snowball branch. There is no `if planet_should_freeze:` anywhere in the code. The bimodal outcome distribution in report §6.2 — temperatures jumping from $\sim 265\text{ K}$ straight to $\sim 230\text{ K}$ with a forbidden gap at $235\text{--}260\text{ K}$ — *emerged* from three lines of albedo parameterisation. The distinction between emergent and imposed behaviour is arguably the most important idea in the entire course, and this is the cleanest demonstration of it available.

Close with honesty (report §9.2): the bifurcation's *existence* is robust EBM physics; its *location* depends on the hard step at T_{ice} and is parameterisation-dependent.

Exercise 4 — Find the cliff

Provide: full single-surface EBM with ice feedback.

Task:

1. Scale S_0 by a factor s . Sweep s downward from 1.0 in steps of 0.01 and record equilibrium global mean T .
2. Identify the critical s at which the planet collapses to snowball.
3. Now sweep *upward* from a snowball state. Does it de-glaciate at the same s ?
4. Plot ice-edge latitude against s for both directions.

Expected result: Steps 3–4 should reveal hysteresis — the classic two-branch structure.

Note for the instructor: report §7.1 found that equilibrium results were robust to initial conditions across 12 runs straddling the bifurcation, so hysteresis in *this* implementation

may be narrower than the textbook picture. That discrepancy is a legitimate discussion topic, not a bug to hide.

Discussion prompt: If Earth had ever entered a snowball state, what could have got it out? (Leads naturally to the carbonate-silicate thermostat, listed as absent in report §9.6.)

ACT II — Orbits and thermal inertia

Lecture 5 — The orbit isn't a circle

Learning objectives

1. Define semi-major axis, eccentricity, obliquity, longitude of perihelion, and explain what each does to climate.
2. Distinguish mean, eccentric, and true anomaly, and explain why all three are needed.
3. Derive or verify the annual-mean insolation identity $\langle S \rangle = S_0 / (4a^2\sqrt{1-e^2})$.
4. Explain why increasing eccentricity *increases* annual-mean insolation.
5. State the Milankovitch hypothesis and the 65 °N June benchmark.

Narrative

Now the orbit earns its place. Introduce the anomalies: M advances linearly in time (so it is what you step), E solves Kepler's equation, v gives actual geometry. Show the Newton iteration and the stable half-angle form for v :

$$v = 2 \cdot \text{atan2}(\beta \cdot \sin(E/2), \cos(E/2)), \quad \beta = \sqrt{(1+e)/(1-e)}$$

Explain why the arccos form is avoided: wrong quadrant, discontinuity through perihelion.

Lead with the counterintuitive result. Annual-mean insolation goes as $1/(a^2\sqrt{1-e^2})$.

Increasing e at fixed a *increases* the annual mean. Students expect the opposite, or expect no change. Kepler's second law does the work: the planet moves fast when close (little time at high flux) and slow when far, and the arithmetic does not cancel.

Then Milankovitch, and the scenario driving this whole project: perihelion in 10%, aphelion out 10%, $e \rightarrow 0.117$, roughly 7× today and 2× the Milankovitch maximum. The benchmark: 65 °N June insolation falls from ~480 to ~400 W/m², a 16.4% drop.

Exercise 5 — Kepler and the annual mean

Provide: `orbital_climate/kepler.py`, `insolation.py`.

Task:

1. Verify the Kepler solver residual $|E - e \cdot \sin E - M| < 10^{-11}$ for $e \in \{0, 0.1, 0.5, 0.9\}$.
2. Round-trip test: $M \rightarrow E \rightarrow v \rightarrow M$ for 1000 random M . Report max error.

3. Numerically integrate daily-mean insolation over one full year **by sampling mean anomaly uniformly**, and compare to $(S_0 / (4a^2 \sqrt{1-e^2}))$.
4. Repeat step 3 sampling *true* anomaly uniformly instead. Explain the discrepancy.
5. Reproduce the Milankovitch benchmark: 65 °N June peak at $e = 0.0167$ and $e = 0.117$.

Expected results: Step 3 → relative error $\sim 2.3 \times 10^{-6}$. Step 4 → substantially wrong, because equal steps in v are not equal steps in time. Step 5 → 478.0 and 399.5 W/m², a -16.4% change.

Why step 4 matters: it is the difference between a test that validates the whole Kepler→insolation chain and a tautology that validates nothing.

Lecture 6 — Time constants (the pivotal lecture)

Learning objectives

1. Define the thermal time constant $\tau = C/B$ and predict its effect on seasonal amplitude.
2. Explain why heat capacity affects the seasonal cycle but not the equilibrium mean.
3. Diagnose why area-weighted averaging of heat capacity fails, in terms of the 175× land/ocean ratio.
4. Derive the $(f/(1-f))$ coupling factor from flux continuity.
5. Explain why land can freeze while ocean at the same latitude does not, and why that matters for glacial inception.

Narrative

Deliver this as a five-act failure, in this exact order. Do not skip to the answer.

Act 1 — the symptom. The model gives a ~10 K seasonal range at 65 °N. Observations over continental interiors give ~40 K. Since seasonal extremes are precisely what drives glacial inception (snow that survives the summer), this is the most load-bearing simplification remaining.

Act 2 — the obvious fix. Make C latitude-dependent as an area blend: $C = f \cdot C_{\text{land}} + (1-f) \cdot C_{\text{ocean}}$. The solver already accepts a C array. Two-line change. Ask the class to predict the outcome.

Act 3 — it fails. At 60 °N (55% land), τ moves from 1.59 yr to 1.44 yr. A 9% change. Essentially nothing.

Act 4 — why. $C_{\text{ocean}} = 2.1 \times 10^8$, $C_{\text{land}} = 1.2 \times 10^6$ J/m²/°C. A factor of 175. The arithmetic mean is the ocean term. Averaging heat capacities is simply the wrong operation on a quantity that governs a response rate. This generalises far beyond climate: **averaging a parameter is not the same as averaging the system's behaviour**, and it fails hardest when the parameter spans orders of magnitude.

Act 5 — the correct formulation (North & Coakley 1979). Two temperatures per latitude, coupled:

$$\begin{aligned}C_l \partial T_l / \partial t &= Q \cdot a(T_l) - (A + B \cdot T_l) + D \nabla^2 T_l + v(T_o - T_l) \\C_o \partial T_o / \partial t &= Q \cdot a(T_o) - (A + B \cdot T_o) + D \nabla^2 T_o - v(T_o - T_l) \cdot f / (1-f)\end{aligned}$$

Before showing the $f/(1-f)$ factor, ask the class to derive it. The flux leaving the ocean column, weighted by ocean area, must equal the flux entering the land column, weighted by land area. Without it the model leaks energy. This is a 90-second derivation that students remember for years.

Then the implementation notes worth mentioning: state becomes a $2N$ vector, the operator a $2N \times 2N$ block matrix, ice-albedo evaluated per surface, land fraction clamped to $[10^{-3}, 1-10^{-3}]$ so $f/(1-f)$ stays finite at Antarctica.

Exercise 6 — Calibrate the coupling

Provide: two-surface EBM with v exposed as a free parameter; the observational targets (~ 40 K land, ~ 8 – 9 K ocean seasonal range at 65°N) but **not** the report's answer.

Task:

1. First reproduce the failure: implement the blended-C approach and measure τ at 60°N . Confirm $\sim 9\%$.
2. Switch to two-surface. Sweep $v \in \{1.0, 2.0, 3.0, 3.5, 4.0, 6.0, 8.0\}$.
3. For each v , record 65°N seasonal range over land and over ocean.
4. Choose the v that best matches both targets simultaneously. Justify the choice.
5. Confirm the global mean temperature barely moves relative to single-surface.

Expected results: They should recover the table in report §4.5 and land on $v = 3.5$ (giving 40.1 K land, 9.1 K ocean). Global mean should shift only from 288.14 $\rightarrow$ 288.07 K.

Why step 5 matters: it confirms Lecture 3's lesson from a new angle. Equilibrium is set by radiation balance, not heat capacity. Report §7.1 records this as a prediction that turned out *wrong* — the author expected to have to re-tune `coalbedo_a0` and did not.

Extension: With v calibrated, rerun the pure eccentricity injection ($a = 1.0$, $e: 0.0167 \rightarrow 0.117$) and measure the 65°N summer peak change for single-surface, two-surface land, and two-surface ocean. Expect -0.04 K, -7.53 K, -0.02 K. The single-surface model misses the glacial-inception signal entirely.

ACT III — From equations to knowledge

Lecture 7 — Turning a PDE into a program

Learning objectives

1. Explain why an equal-area grid in $\sin \phi$ eliminates a class of weighting bugs by construction.
2. Distinguish explicit, implicit, and IMEX time stepping and state when each is appropriate.
3. Explain what "stiff" means and why the diffusion + OLR terms qualify.
4. Verify that the discrete diffusion operator conserves energy exactly.
5. Articulate the difference between a timestep chosen for accuracy and one forced by stability.

Narrative

Only now show the complete governing equation:

$$C \partial T / \partial t = Q(x, t) \cdot a(x, T) - (A + B \cdot T) + D \partial / \partial x [(1 - x^2) \partial T / \partial x]$$

Every term is now familiar. Ask the class to name each one's origin lecture before proceeding.

Grid design. Equal-area cells in $x = \sin \phi$. Because cells have equal area, the global mean is the plain cell mean — no $\cos \phi$ weighting is required, and therefore no $\cos \phi$ weighting bug is *possible*. Contrast with a uniform-in-latitude grid where every global-mean computation is a place to make the same mistake.

Conservation by construction. The flux-divergence operator uses weight $w = 1 - x^2$ at cell interfaces. Since $w = 0$ at both poles, no-flux boundaries are automatic and the operator's area-weighted sum vanishes identically for *any* field. Energy conservation is a structural property of the discretisation, not something to be hoped for and checked.

IMEX splitting. Linear diffusion and OLR treated implicitly; the nonlinear $Q \cdot a(T)$ source explicitly at the old temperature:

$$\begin{aligned} M_{\text{imp}} &= \text{diag}(C/\Delta t + B) - D \cdot L && \text{(LU-factorised once, reused every step)} \\ \text{rhs} &= (C/\Delta t) \cdot T + Q \cdot a(T) - A \end{aligned}$$

The payoff is that dt_days becomes an accuracy knob, not a stability limit. The report verifies consistency from $\text{dt} = 1$ day to $\text{dt} = 30$ days. Emphasise how much this buys: a fully explicit scheme with $n_{\text{lat}} = 180$ would need a punishingly small dt for the diffusion term alone.

Exercise 7 — Break the stepper

Provide: EBM with the time-stepping scheme swappable between explicit and IMEX.

Task:

1. With IMEX, run identical scenarios at $\Delta t = 1, 5, 10, 30$ days. Compare equilibrium global means and seasonal ranges.
2. Switch to fully explicit. Find the largest Δt that remains stable at $n_{\text{lat}} = 180$.
3. Repeat step 2 at $n_{\text{lat}} = 45$ and $n_{\text{lat}} = 90$. Deduce the scaling of the stability limit with grid resolution.
4. Verify conservation directly: construct a random temperature field, apply the diffusion operator, compute the area-weighted sum.
5. Verify the operator annihilates a constant field.

Expected results: Step 1 $\rightarrow$ results agree closely; differences are accuracy, not instability. Step 3 $\rightarrow$ stability limit scales as roughly Δx^2 , i.e. as $1/n_{\text{lat}}^2$. Step 4 $\rightarrow$ sum $\sim 10^{-12}$. Step 5 $\rightarrow$ exactly zero to machine precision.

Lecture 8 — How do you know the model isn't lying to you?

Learning objectives

1. Rank validation tests by strength and explain the ranking.
2. Construct an analytic limit test for a numerical model.
3. Explain why a test asserting a hardcoded constant can be worse than no test.
4. Distinguish tests that check internal consistency from tests that check correctness.

Narrative

This deserves a full lecture and almost never gets one. Present a hierarchy, strongest first:

Tier 1 — analytic limits. Disable a mechanism until an exact answer exists, then demand it. The strongest example in this codebase: set $D = 0$ and suppress ice feedback, and every latitude must independently reach $\overline{T} = (\bar{Q} \cdot a_0 - A)/B$. It matches to 0.05°C . This single test exercises the grid, the insolation chain, the stepper, and the radiation terms against a known closed-form answer.

Tier 2 — conservation and identities. Diffusion annihilates a constant (exactly). Area-weighted operator sum vanishes ($\sim 10^{-12}$). Global energy budget closes: $\overline{(\text{absorbed})} = A + \overline{B(T)}$ to 0.05 W/m^2 . Round-trip $M \rightarrow E \rightarrow v \rightarrow M$.

Tier 3 — physical benchmarks. Present-day global mean 288.15 K . Land fraction integrating to 0.290 against Earth's 0.29 . Perihelion date falling on Jan 1, consistent with $\lambda_p \approx 282^\circ$.

Tier 4 — internal consistency. Zero variance in pre-flyby values across 672 runs. Bound count 662/672 matching an independent source. These catch pipeline bugs but cannot catch physics errors.

Then the punchline. Report §7.1: a Kepler test asserted a hardcoded constant of `1.1934205`. Recomputed independently, the correct value is `1.1853242`. **The test would have certified a broken solver.** A test is code; code has bugs; a test whose expected value was produced by the thing it tests is a tautology wearing a lab coat.

Close with the bug catalogue (§8) as a taxonomy — a regex assuming XML attribute order, a double-blend causing a shape error, a `--limit` flag truncating before statistics were computed and reporting "654 outside validity band" when only 8 runs were requested. That last one is worth dwelling on: it produced a *plausible-looking wrong number*, which is the most dangerous class of bug.

Exercise 8 — Write the tests, then break the code

Provide: the EBM package, and a copy with three deliberately injected bugs (suggested: a sign error in the diffusion flux; a $\cos \phi$ weighting applied to an already-equal-area grid; an off-by-one in the seasonal insolation index).

Task:

1. Before looking at the injected-bug version, write four tests: one analytic-limit, one conservation, one round-trip, one physical benchmark.
2. Run your tests against the buggy version. How many bugs does your suite catch?
3. For any bug you missed, write the test that would have caught it.
4. Take one existing test in `orbital_climate/tests/` and determine whether its expected value was derived independently or is a tautology.

Assessment note: step 2 is the graded item, and a low catch rate is a legitimate outcome to discuss rather than penalise. Most students catch the sign error and miss the off-by-one.

Lecture 9 — Spurious signals and honest limits

Learning objectives

1. Identify collinearity as a source of spurious correlation and control for it.
2. Use physical reasoning to predict what a statistical analysis *must* show, and use that as a check.
3. Explain regime dependence: why a more complete model is not uniformly "more" or "less" of anything.
4. Quantify where a model's parameterisations become invalid and state results as ordinal rather than quantitative outside that range.

Narrative

Part 1 — the spurious correlation. Report §6.3. Across the sweep, λ_p correlates with temperature at +0.593. Substantial-looking. But λ_p is 42% correlated with semi-major axis a in this sweep. Control for annual-mean insolation and the λ_p correlation collapses to -0.22 , while (S_{mean}) alone explains +0.998.

What makes this the best confounding example available in a physical-science context: **the physics tells you the answer in advance.** λ_p shifts seasonal *phasing*, not the annual mean. It therefore cannot have a large direct effect on equilibrium temperature. The residual -0.22 is the genuine, small, seasonal/ice-albedo effect. The statistics were checkable against theory rather than merely trusted.

Part 2 — regime dependence. Subtler and more important. The two-surface model is unambiguously more physical. Yet across all 220 habitable-band runs it shows *less* summer cooling than the single-surface model (median -3.37 K vs -6.31 K; 47% vs 63% of runs cooling more than 4 K).

Why: in this sweep a ranges from 0.63 to 2.8 AU, so annual-mean insolation changes dominate. A heavily damped single-surface world has summer $\approx$ annual mean, so it inherits the full annual-mean cooling. Land, which sits well above the annual mean in summer, is partly buffered.

Now isolate the mechanism: restrict to the 72 runs where a stayed within 2% of 1.0 AU, so only orbital *shape* changed. Single-surface reports +0.48 K summer *warming*; two-surface land reports -0.98 K *cooling*. **A sign flip.**

The accurate statement — worth having students write out in their own words — is that the two-surface model is correct in both regimes, but the direction of the correction it applies depends on what changed.

Part 3 — honest limits. Report §9.1. The linear OLR is calibrated near 288 K; only 33% of runs land in the defensible 250–300 K band. At the 559.7 K extreme, $(A + B \cdot T)$ implies roughly 800 W/m² outgoing where blackbody physics demands ~ 5500 W/m². The model returns a confident, precise, entirely fictitious number. Connect back to Lecture 1, where this was flagged in advance.

Exercise 9 — Find the confounder yourself

Provide: $(\text{<STAMP>_climate.csv})$ and $(\text{<STAMP>_earth_elements.csv})$ (or synthetic equivalents), with no guidance as to which predictors matter.

Task:

1. Compute pairwise correlations between equilibrium T and each of a , e , ϵ , λ_p .
2. Rank them. Write down, before any further analysis, which you believe are causal.

3. Compute the pairwise correlation matrix among the *predictors*. Identify collinear pairs.
4. Construct the derived predictor $\left(S_o / (4a^2 \sqrt{1-e^2})\right)$ and correlate it with T.
5. Partial out annual-mean insolation and recompute the λ_p correlation.
6. From physics alone, predict what step 5 should give, and compare.

Expected results: Step 1 $\rightarrow$ roughly +0.966 (S_{mean}), -0.790 (a), +0.593 (λ_p), +0.110 (e), +0.067 (ϵ). Step 5 $\rightarrow$ -0.22. Step 6 is the graded item: the physical prediction (λ_p affects phasing, not annual mean, so near-zero with a small seasonal residual) should precede and validate the statistics.

Extension: Restrict to runs with $|a - 1.0| < 2\%$ and reproduce the sign flip in summer-peak change.

Lecture 10 — The full pipeline (capstone)

Learning objectives

1. Explain how orbital elements were recovered from a simulation whose output did not contain them.
2. Describe frame identification by testing rather than by assumption.
3. Explain why a Cartesian-product sweep is the wrong tool for paired parameters.
4. Interpret the sweep's headline results, including the transient finding.

Narrative

The gap. REBOUND runs write a , e and q before and after the flyby — the orbit's size and shape. A seasonal climate model also needs its *orientation*: λ_p (which hemisphere gets summer at perihelion) and obliquity. Neither is in the file.

Note also the configuration trap: a YAML `sweep:` block computes a Cartesian product, but these parameters are *paired* — run N's (a , e , ϵ , λ_p) belong together. Sweeping 662 values of a against 662 of e yields 438,244 physically meaningless combinations. Hence a bridge script rather than configuration. Worth pausing on: this is a case where the available abstraction actively invited a wrong answer.

The detective story (§5.2), presented as a chain of inference:

1. Test, don't assume: at $t = 0$, Earth's orbit is inclined 23.457° to the z -axis.
2. That is exactly the obliquity. Therefore the simulation runs in the equatorial J2000 frame.
3. Therefore the frame's z -axis *is* Earth's spin axis.
4. A point-mass integration applies no torque to that axis, so it stays fixed in inertial space while the black hole tilts the orbital plane.

5. Therefore: $\varepsilon = \text{angle}(\hat{z}, \hat{h})$; equinox direction $\hat{e} = \hat{h} \times \hat{s} \perp$; λ_p = angle from $\hat{e}$ to the eccentricity vector about $\hat{h}$.

Validation (§5.3): applied at $t = 0$ the recovery must return present-day Earth, and does — $a = 0.99981$ AU, $e = 0.01717$, $\varepsilon = 23.457^\circ$, $\lambda_p = 282.29^\circ$ against $\sim 283^\circ$ expected. Three independent cross-checks: perihelion falling on Jan 1 (consistent with $\lambda_p \approx 282^\circ$), zero variance across all 672 runs, and bound count matching an independent source exactly.

Honest approximation (§5.4): Earth's spin axis precesses on a 25,772-year cycle, so it was $\sim 0.70^\circ$ from $\hat{z}_{J2000}$ in 1873 and $\sim 1.01^\circ$ the other way by 2181. True λ_p drifts $\sim 4.3^\circ$ over the run. Negligible here because the induced λ_p spread is the full $0-360^\circ$, so 4.3° is $\sim 1\%$ of the signal — and because REBOUND treats Earth as a point mass, so precession cannot be modelled regardless of frame choice.

Results (§6). Baseline 288.19 K; median outcome 278.97 K (-9.2 K); 21% recognisably Earth-like; 39% snowball. The bimodal distribution with its forbidden gap. The overlap zone at $a = 1.073-1.122$ AU where both outcomes occur depending on other parameters.

The finding to end on (§6.5): 198 runs (31%) show peak hemispheric asymmetry above 5 K decaying to under 1 K, typically peaking in year 1. Nearly a third of the sweep passes through a substantially lopsided climate that leaves *no trace whatsoever* in the equilibrium state. The equilibrium sweep structurally could not have produced this. Choosing what to compute determines what you are capable of discovering.

Exercise 10 — Capstone

Task (choose one, written up as a short report with figures):

(a) Recovery. Given a synthetic orbit with known ε and λ_p , implement the frame-based recovery and demonstrate it returns the known values. Then perturb the frame assumption by 1° and quantify the error induced in λ_p .

(b) Regime map. Using the sweep outputs, produce a two-dimensional map in (a, e) coloured by outcome class (temperate / snowball / outside validity band). Identify and characterise the overlap zone. Determine what distinguishes temperate from snowball outcomes *within* it.

(c) Transient hunt. Identify the runs with the largest peak hemispheric asymmetry. Determine what orbital configuration produces them, and explain the mechanism. Verify that the asymmetry decays to zero at equilibrium.

(d) Break the model honestly. Choose one limitation from §9 and quantify it. E.g. replace linear OLR with a grey-body (σT^4) form and re-run the sweep; report how many outcome classifications change, and in which direction.

Assessment structure

Component	Weight	Notes
Lab exercises 1-5	20%	Completion-graded; these build the tooling
Exercise 6 (calibration)	15%	The key skill: calibrating against observation
Exercise 8 (testing)	15%	Graded on the reasoning in steps 2-3, not the catch rate
Exercise 9 (confounding)	20%	Step 6 — physical prediction before statistics — is the core
Capstone	30%	Written report with figures

Rationale for the weighting. Exercises 6, 8 and 9 carry disproportionate weight because they assess judgement rather than execution. Anyone can run a model; the course is about knowing when to disbelieve one.

Compression options

Eight sessions: merge L1 + L2 (energy balance and geometry in one), and deliver L7 as a demonstration rather than a lab.

Six sessions: additionally merge L3 + L4 (transport and feedback together), and fold L5 into L6 with Kepler as pre-reading.

Do not cut L6 or L9 under any circumstances. The heat-capacity failure and the spurious correlation are the two places where this course teaches something a student cannot get from a standard textbook, and the two places where this source material is genuinely distinctive. A course that keeps L6 and L9 while dropping half of Act I is better than the reverse.

Instructor notes

On the negative results. Report §7.2 lists genuine negative findings: temperature overshoot is exactly zero across all 644 runs in both modes; initial-condition bistability does not occur in the tested range; the blended heat capacity is not a useful approximation. Distribute these deliberately — students rarely see negative results presented as findings rather than as absences, and the habit of recording "we checked, and it doesn't happen" is worth transmitting.

On the report as an artefact. Consider assigning §7 as a standalone reading in week one, before any physics. A document that opens by listing which of its author's initial assertions turned out to be wrong models an intellectual posture more effectively than any amount of exhortation.

On what is deliberately absent. No carbonate-silicate thermostat, no clouds, no ocean heat transport beyond diffusion, no precession or obliquity cycling within a run, zonal-mean geometry only. Each of these is a natural extension project and a natural exam question of the form "what would you need to add, and what would change?"